

3. Emerging Digital Assets

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C O N T E N T S

1. Overview of Crypto Assets

Crypto Related Financial
Products

Digital Assets

2. Crypto Derivatives

ETF

Crypto Futures

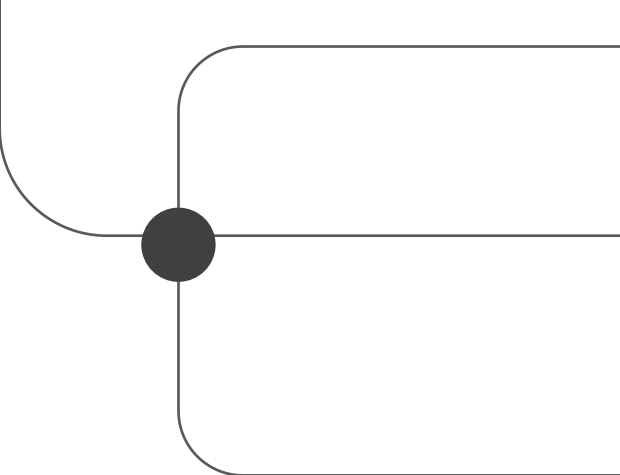
Crypto Options

3. Alternative Crypto Markets

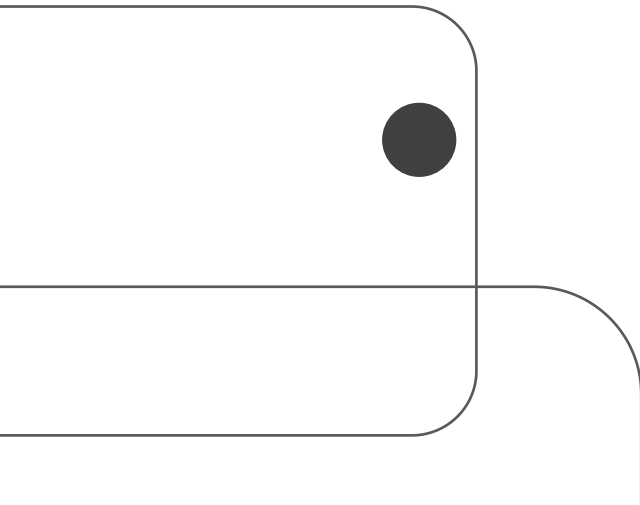
Polymarket

NFTs

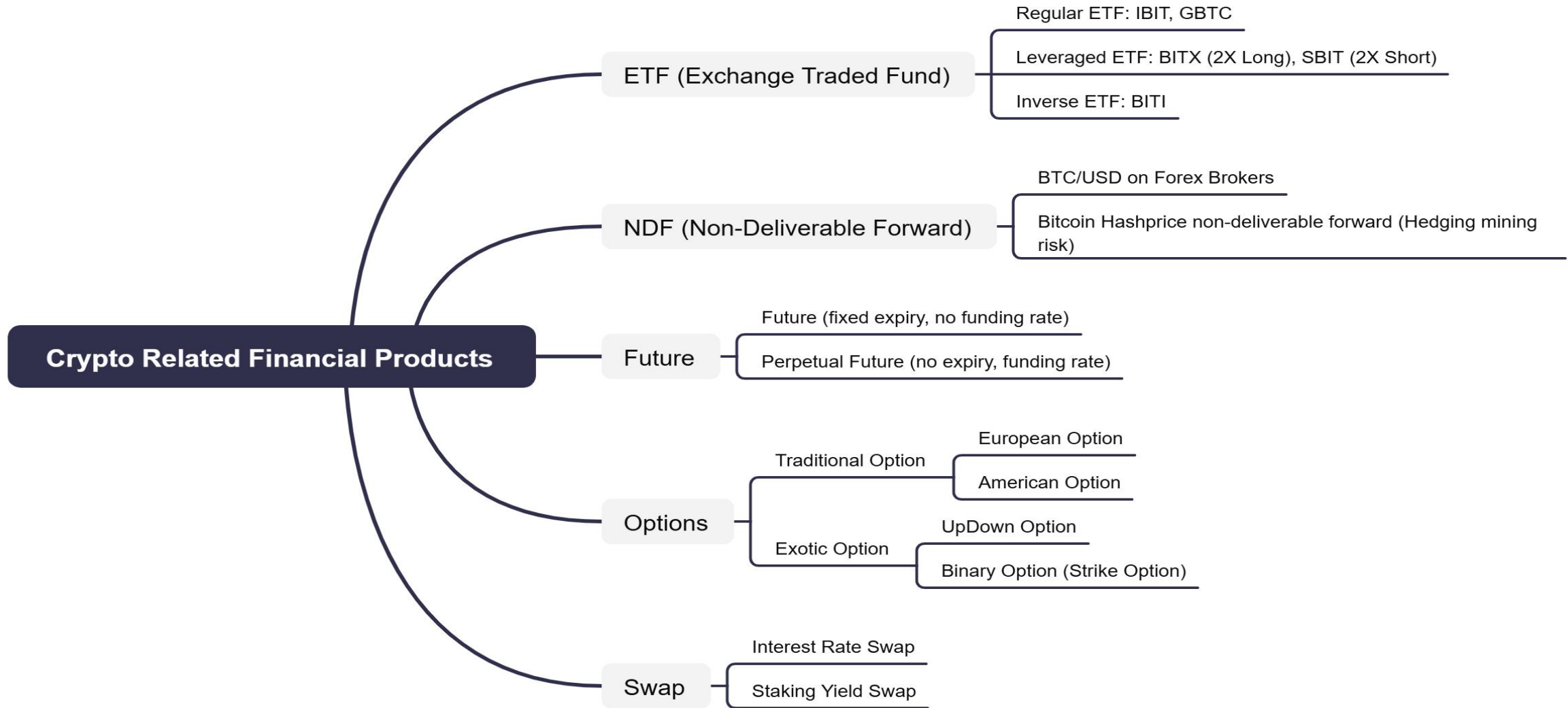
GameFi



1. Overview of Crypto Assets

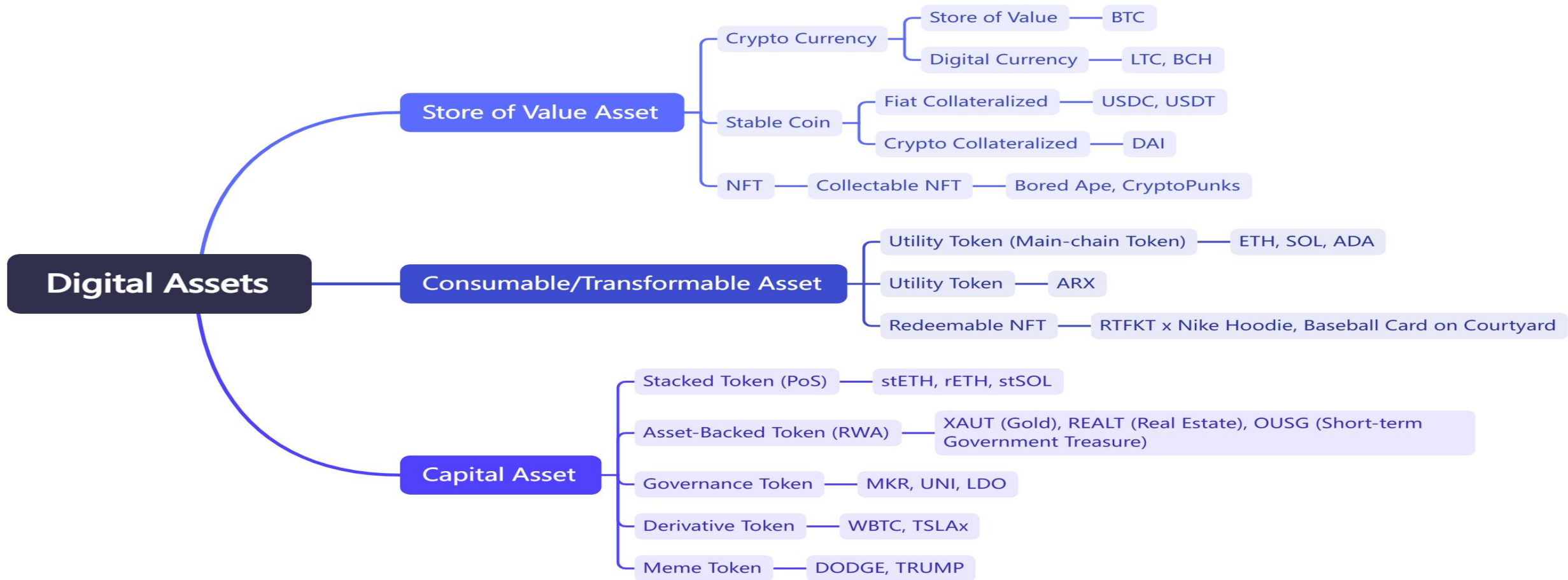


1. Overview of Crypto Assets: Crypto Related Financial Products

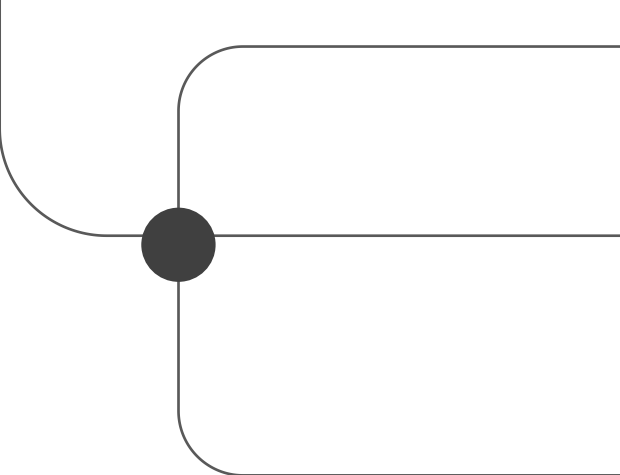


- Regulated by SEC, CFTC, etc.
- Available through major exchanges such as NYSE, ICE, CME, etc.

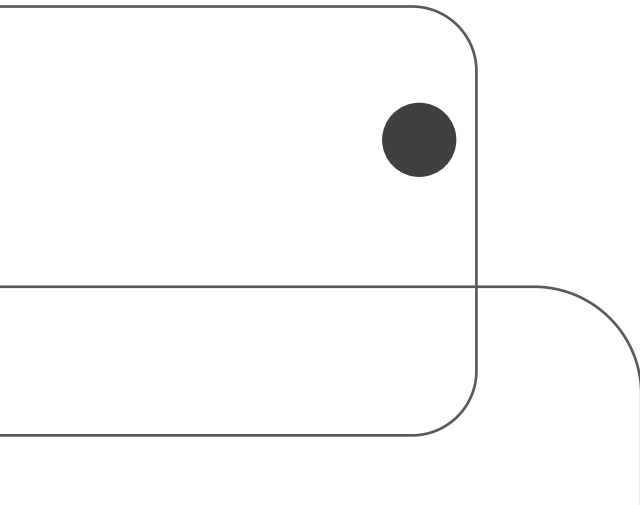
1. Overview of Crypto Assets: Digital Assets

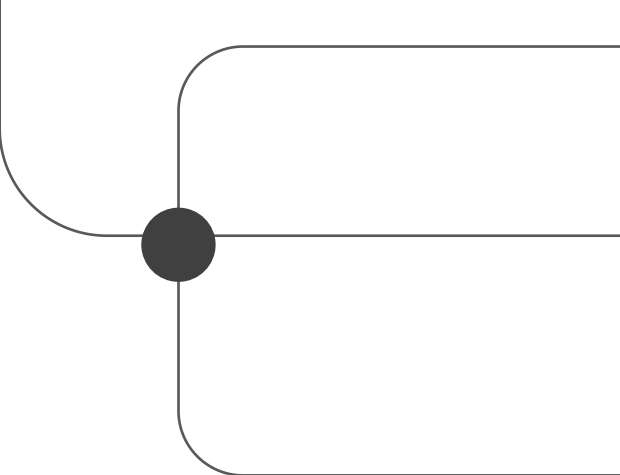


- Most of the crypto currencies are not complianced with security law.
- Available through crypto exchanges such as Coinbase, Binance, Uniswap, etc.
- Crypto drivatives are available Crypto.com, BitMex, Binancie Options, Deribit,etc.
- NFT are available on marketplaces such as OpenSea, Courtyard.io, etc.

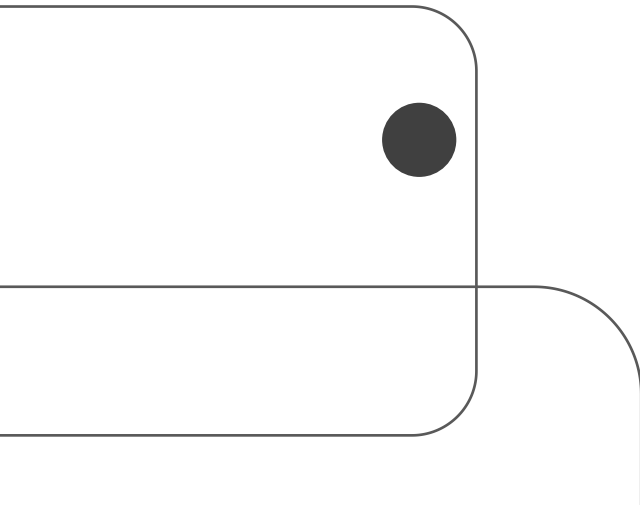


2. Crypto Derivatives

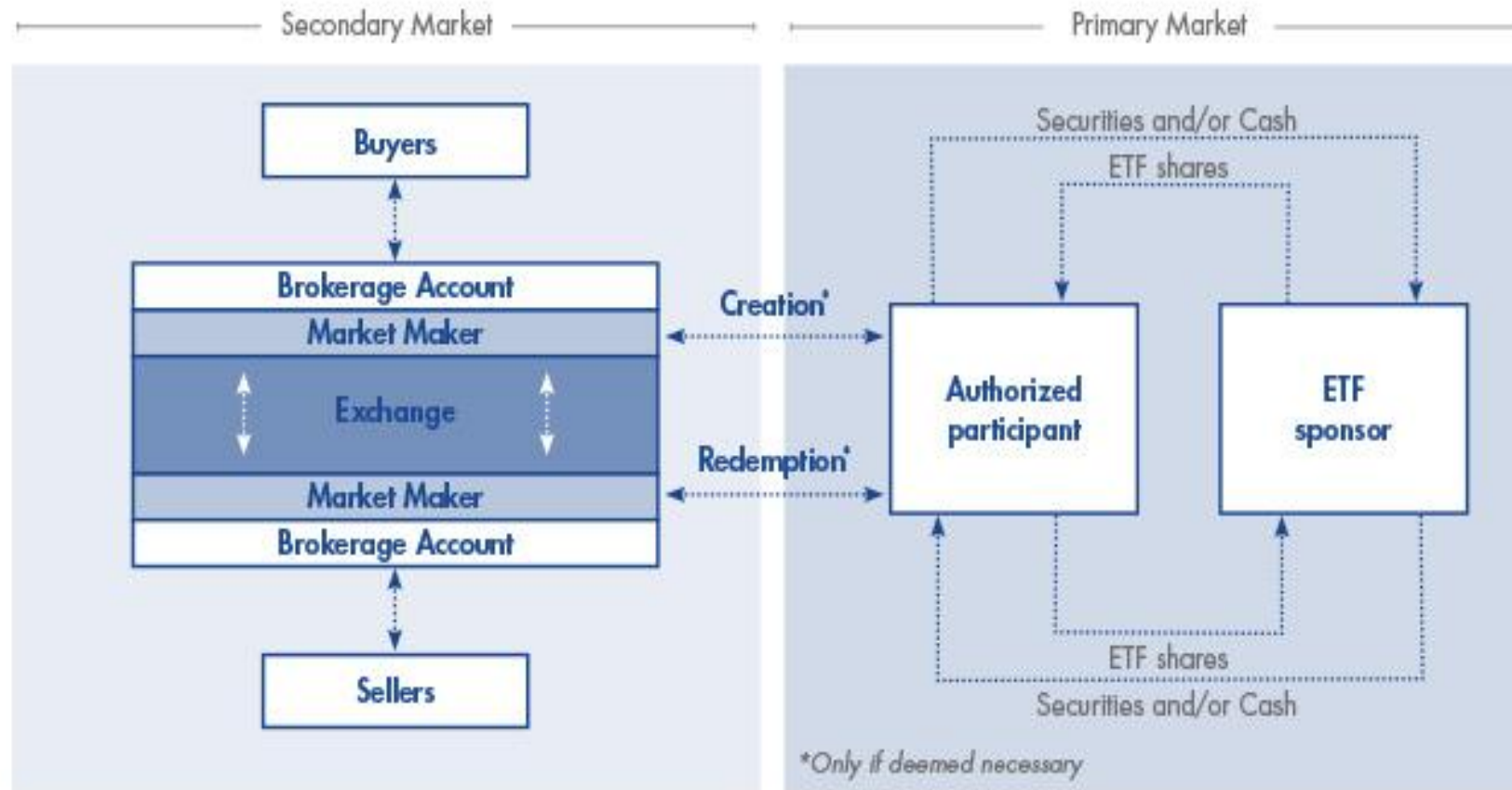




ETF and Leveraged ETF

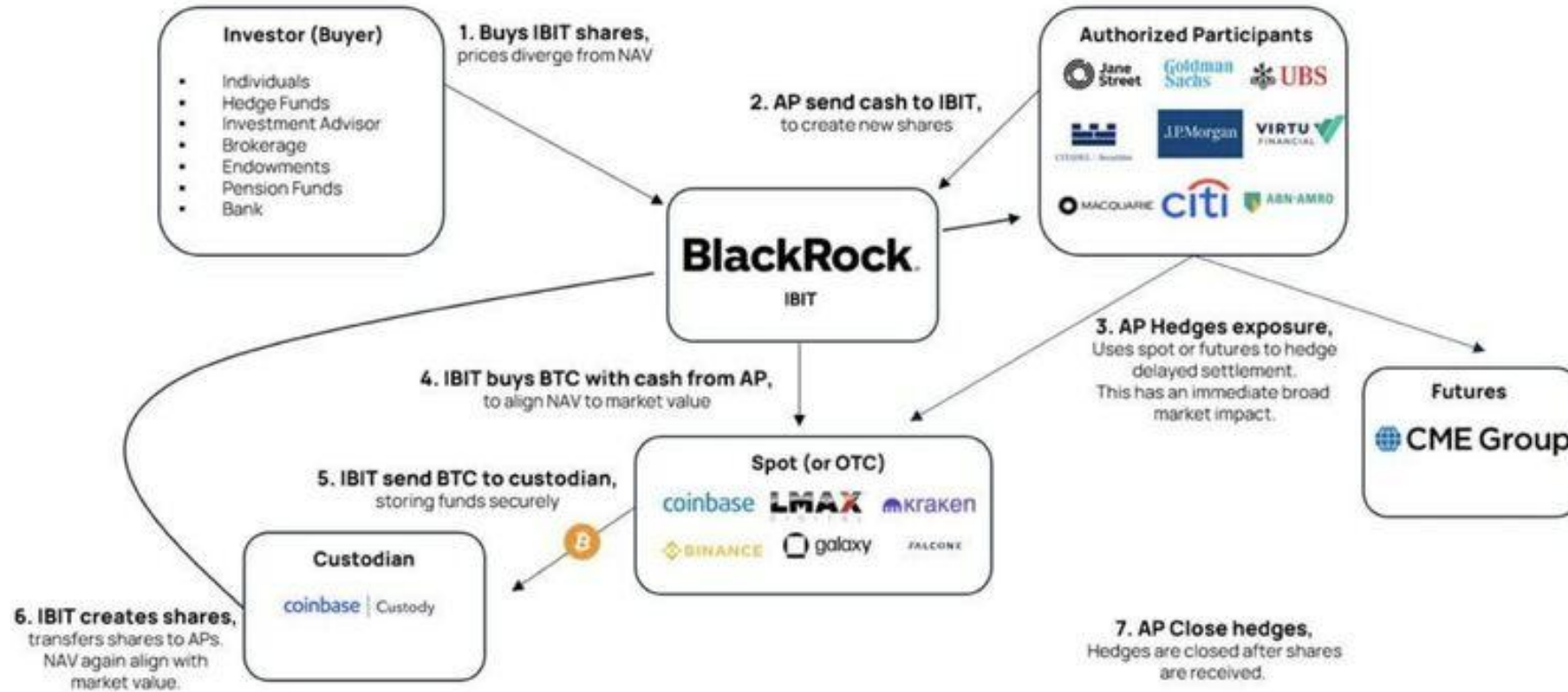


2. Crypto Derivatives: Exchange Traded Fund (ETF) : Creation/Redemption/Circulation



- ETF can be created or redeemed through primary market.
- Investors can buy or sell ETFs through secondary market

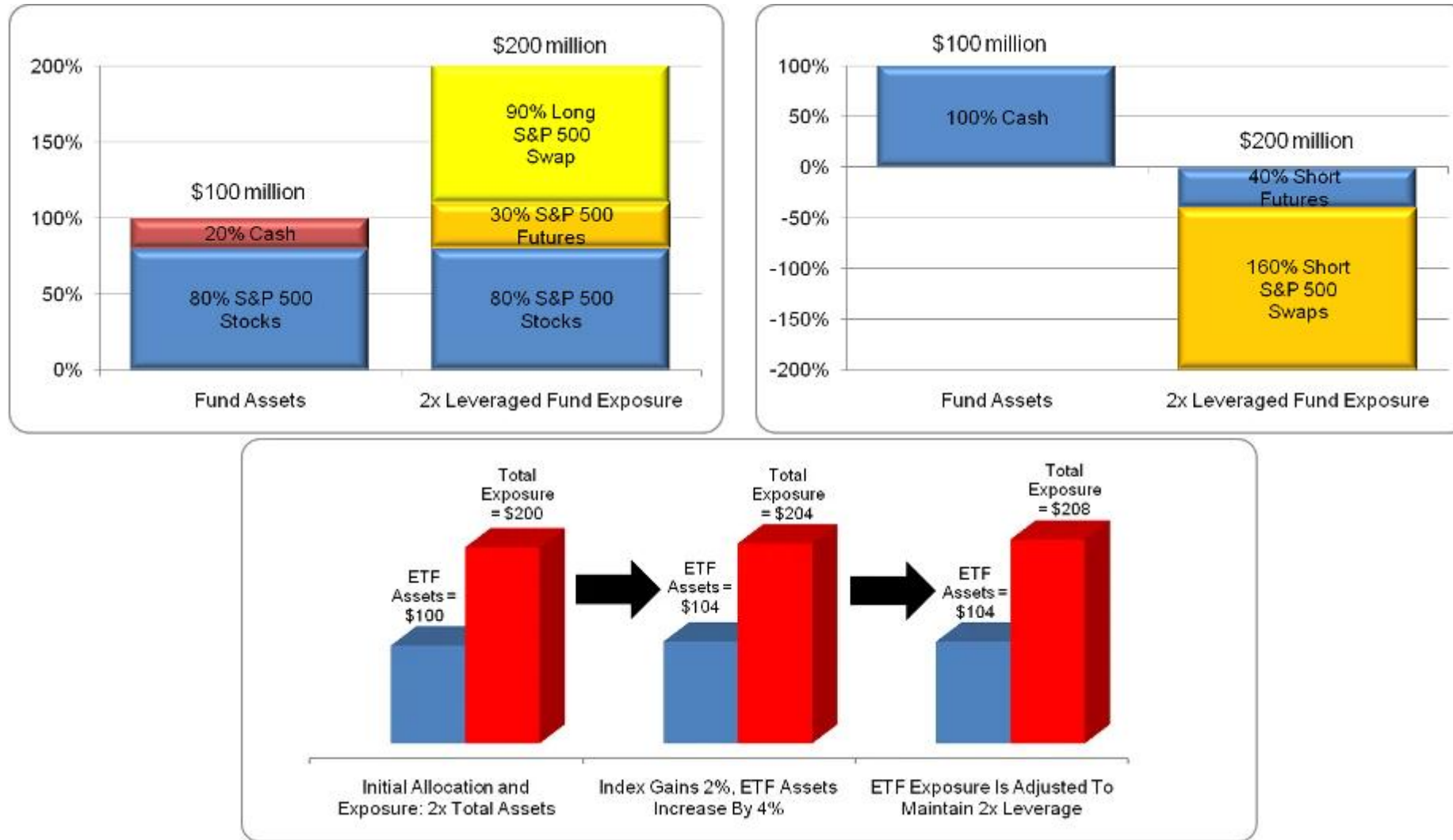
2. Crypto Derivatives: Exchange Traded Fund (ETF) : IBIT Creation Process



source: Oliver Yates

- This figure shows the creation process of IBIT.
- The NAV of ETF and spot BTC price may differ during the delays in creation/redemption.
- ETFs may have tracking errors from their underlying assets due to trading behaviors.

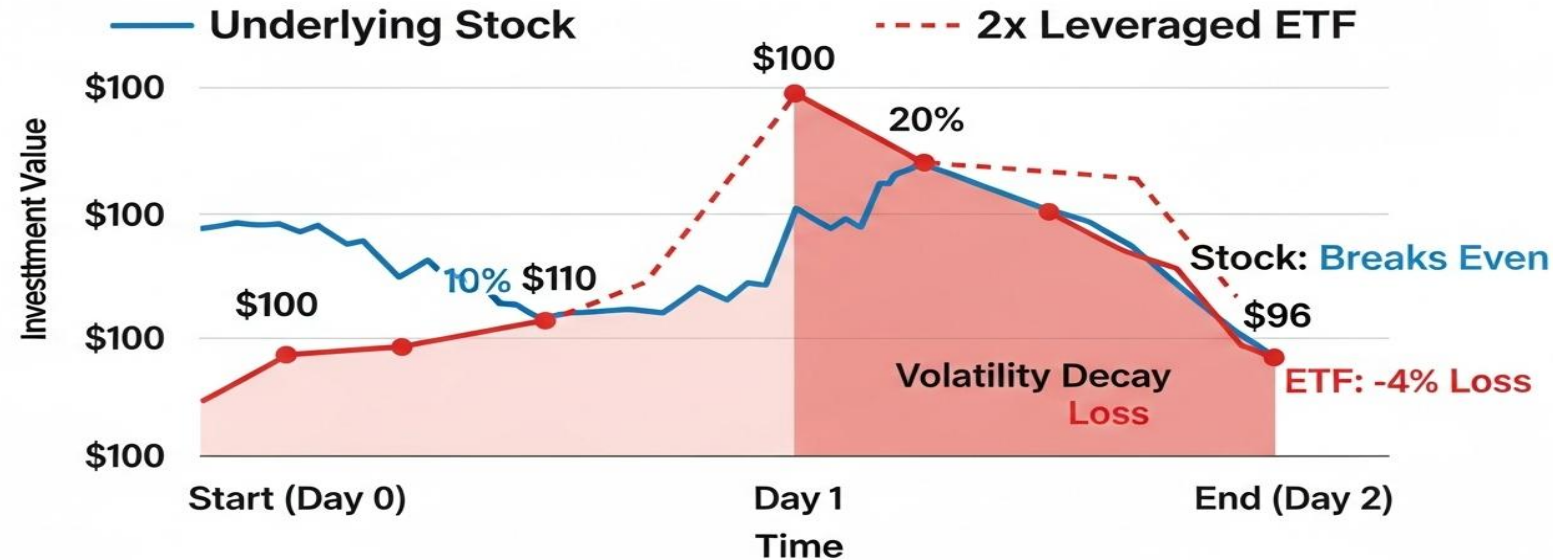
2. Crypto Derivatives: Exchange Traded Fund (ETF) : Leveraged ETF, inverse ETF



- Leveraged ETFs rebalance daily, not designed for long-term holding.

2. Crypto Derivatives: Exchange Traded Fund (ETF) : Leveraged ETF, Impact of Volatility Decay

The Impact of Volatility Decay



- **Volatility Decay:** Leveraged ETFs rebalance daily, leading to compounding effects.

e.g.

Day 1: The underlying stock rises 10% from \$100 to \$110.

The 2x leveraged ETF rises 20%, from \$100 to \$120.

Day 2: The underlying stock falls 9.09% from \$110 back to \$100.

The 2x leveraged ETF falls 18.18% (2 x 9.09%). Its new value is $\$120 - (18.18\% \text{ of } \$120) = \$98.18$.

2. Crypto Derivatives: Exchange Traded Fund (ETF) : Observations

- Increases in ETF ownership led to significant decreases in stock price volatility and greater market liquidity.
- ETFs increase the volatility of their underlying symbols.
- Purchases and sales of commodities by ETFs can significantly affect the price of such commodities.
- When the dividends are negatively correlated, ETFs reduce mispricing and price volatility without decreasing trading volume.
- ETFs have significantly lower tax burdens compared to mutual funds.
- ETF ownership of stocks leads to higher volatility and turnover.
- ETFs can contribute to systemic risk by inducing investors to take correlated exposures that may trigger a chain reaction.
- ETFs could increase asset volatility and correlations across baskets that have similar systematic factors.

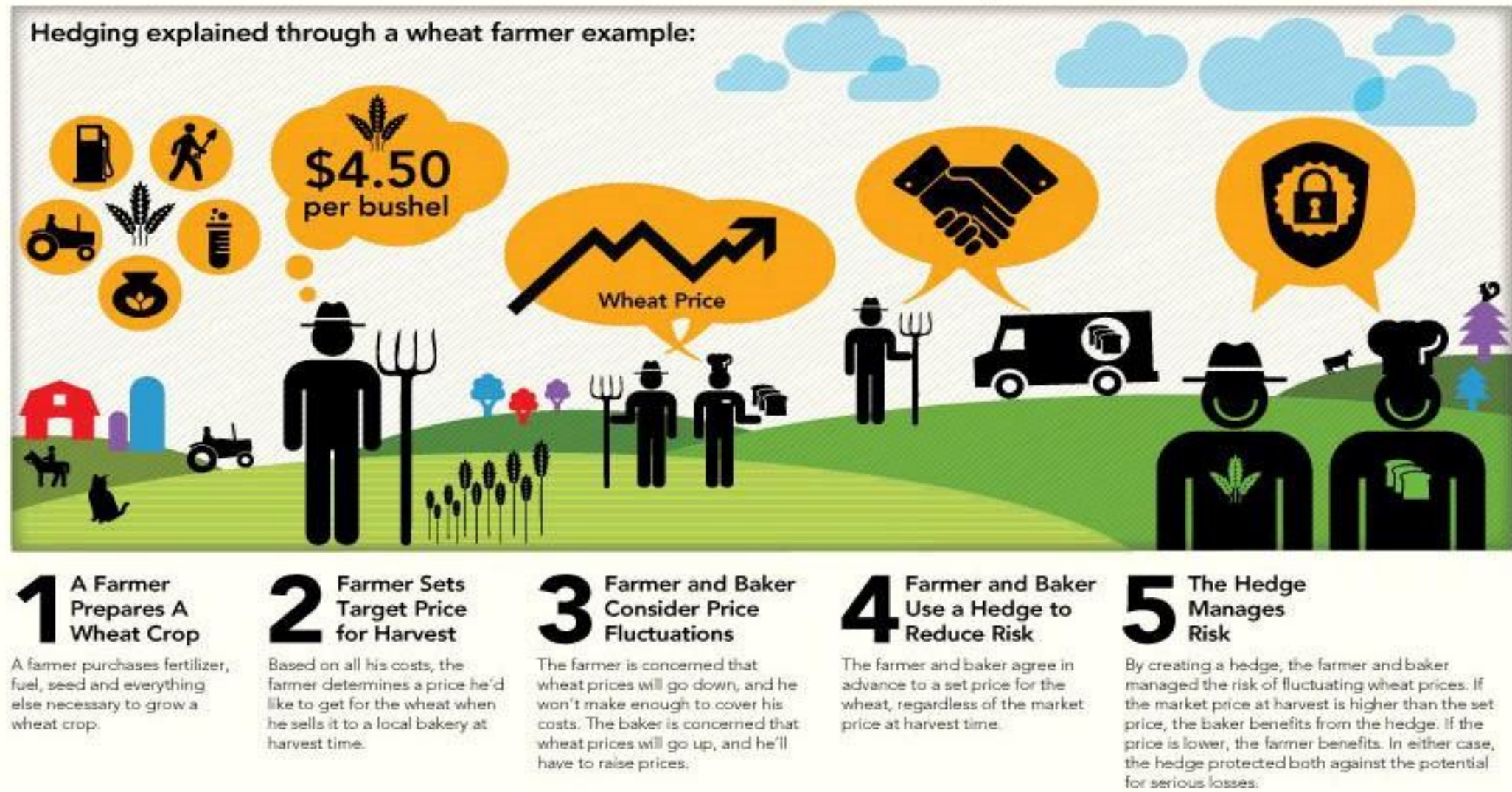
2. Crypto Derivatives: Exchange Traded Fund (ETF) : A Partial List of Crypto ETFs

Symbol	Fund Name	Stock Price	% Change	Assets ~
IBIT	iShares Bitcoin Trust ETF	51.20	-1.73%	71.03B
FBTC	Fidelity Wise Origin Bitcoin Fund	78.67	-1.64%	18.50B
GBTC	Grayscale Bitcoin Trust ETF	70.58	-1.69%	15.44B
BTC	Grayscale Bitcoin Mini Trust ETF	39.95	-1.72%	4.53B
BITB	Bitwise Bitcoin ETF Trust	49.04	-1.70%	3.69B
ARKB	ARK 21Shares Bitcoin ETF	29.95	-1.71%	3.56B
BITO	ProShares Bitcoin ETF	13.33	-1.62%	2.46B
BITX	2x Bitcoin Strategy ETF	30.72	-3.43%	1.66B
HODL	VanEck Bitcoin ETF	25.53	-1.62%	1.53B
BITU	ProShares Ultra Bitcoin ETF	29.94	-3.48%	787.86M
BTCO	Invesco Galaxy Bitcoin ETF	89.95	-1.73%	581.86M
GDLC	Grayscale CoinDesk Crypto 5 ETF	42.91	-1.99%	574.05M
EZBC	Franklin Bitcoin ETF	52.19	-1.79%	533.85M
BRRR	Coinshares Bitcoin ETF	25.49	-1.75%	526.87M
MSTX	Defiance Daily Target 2x Long MSTR ...	5.35	-7.92%	397.09M
WGMI	CoinShares Bitcoin Mining ETF	42.74	-7.77%	270.91M
YBTC	Roundhill Bitcoin Covered Call Strate...	31.62	-1.73%	245.28M
BTCW	WisdomTree Bitcoin Fund	95.61	-1.70%	146.01M
BITI	ProShares Short Bitcoin ETF	22.55	1.67%	112.13M
SBIT	ProShares UltraShort Bitcoin ETF	41.69	3.53%	104.10M

source: <https://stockanalysis.com/list/bitcoin-etfs/>

Future

2. Crypto Derivatives: Futures



2. Crypto Derivatives: Futures, Observations



2. Crypto Derivatives: Futures

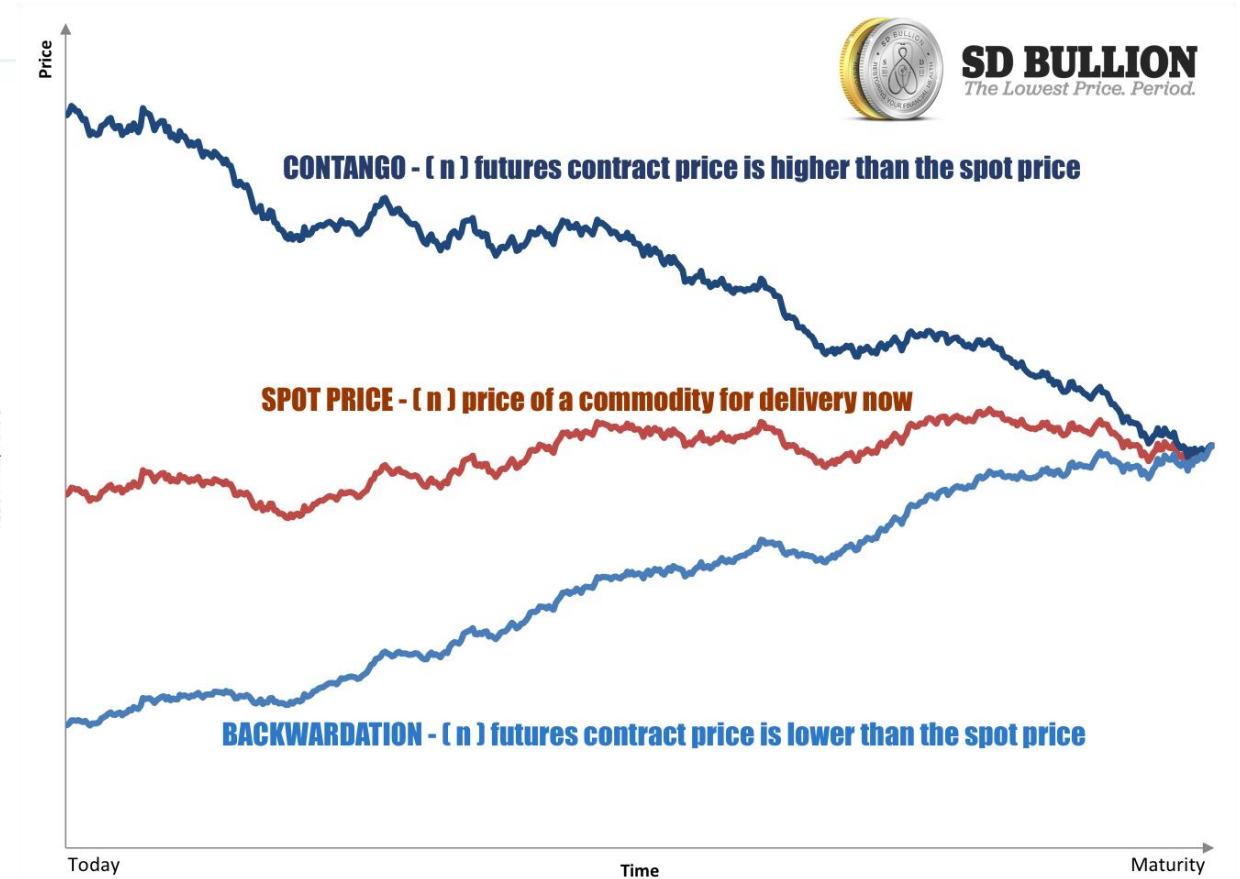
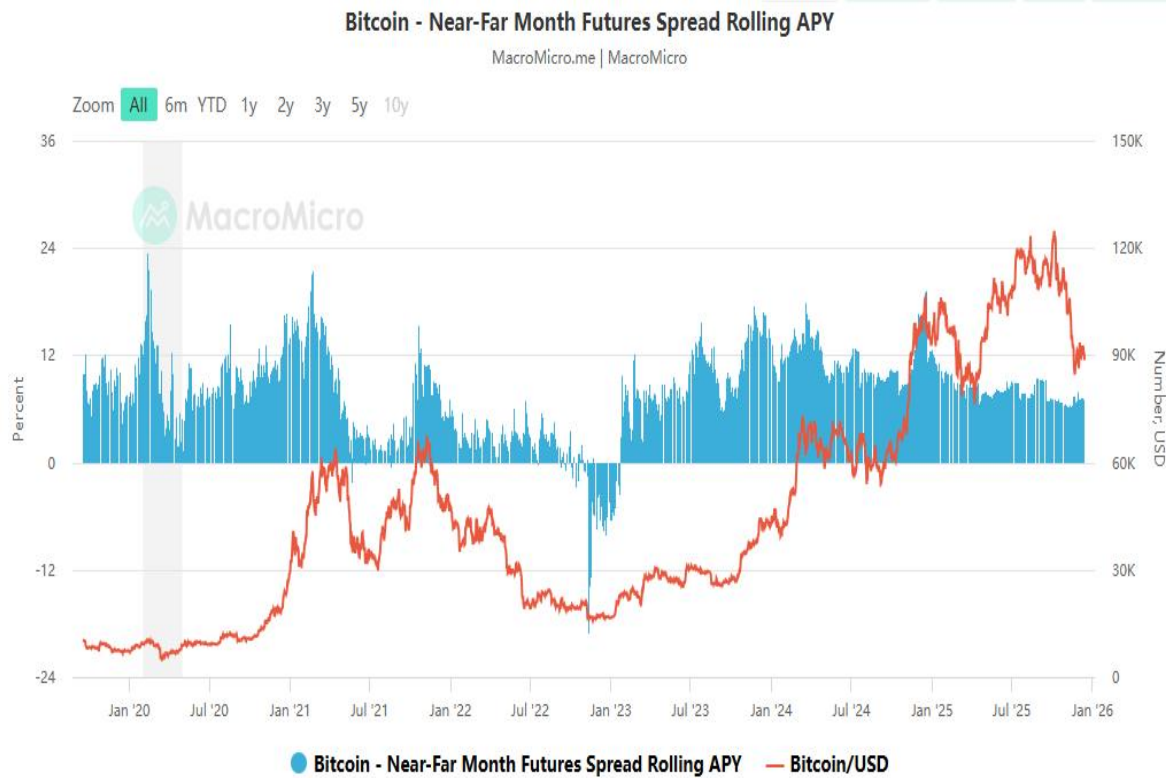
BITCOIN FUTURES - SETTLEMENTS

TRADE DATE Friday, 12 Dec 2025 ▾

Last Updated 12 Dec 2025 11:55:00 PM CT

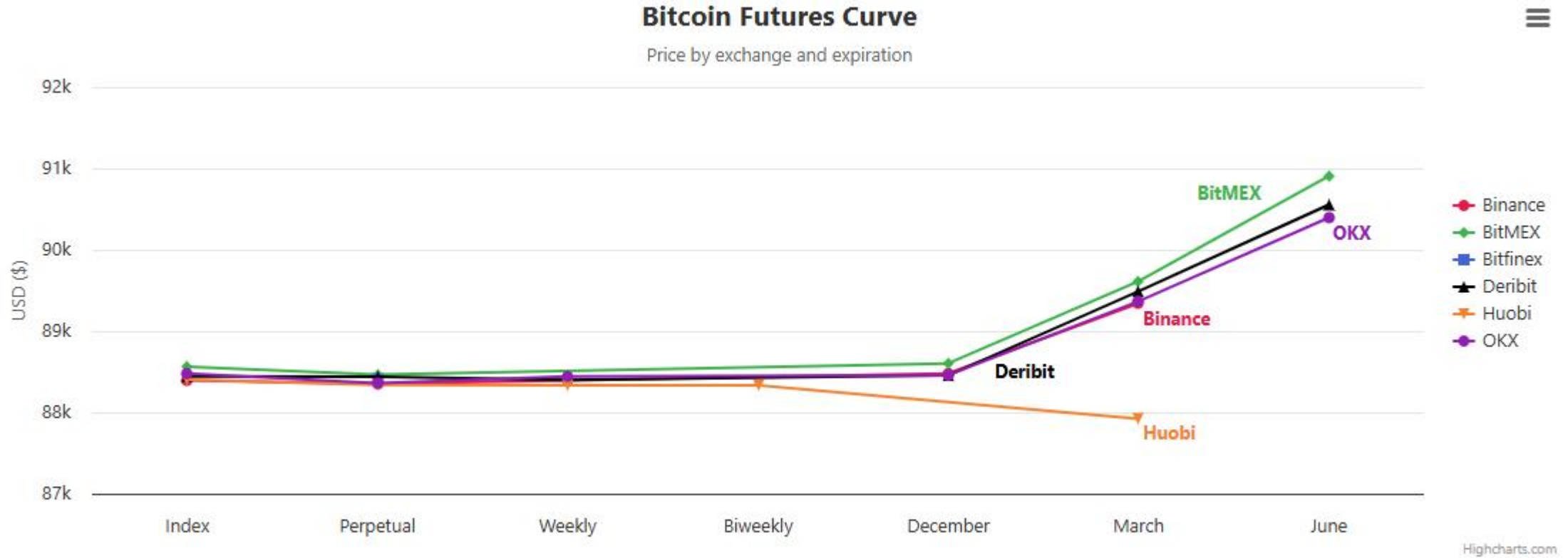
ESTIMATED VOLUME TOTALS 8,001					PRIOR DAY OPEN INTEREST TOTALS 24,692			
MONTH	OPEN	HIGH	LOW	LAST	CHANGE	SETTLE	EST. VOLUME	PRIOR DAY OI
DEC 25	92725.00	93095.00B	89640.00	90330.00	-1570.00	90440.00	6,577	18,217
JAN 26	93330.00	93635.00B	90180.00A	90825.00	-1585.00	90960.00	1,405	6,101
FEB 26	93900.00	94005.00B	90640.00A	91695.00A	-1595.00	91385.00	19	217
MAR 26	-	94400.00B	91125.00A	94400.00B	-1610.00	91820.00	0	144
APR 26	-	94790.00B	91665.00A	94790.00B	-1625.00	92320.00	0	8

2. Crypto Derivatives: Futures, Near-Far Month Spread



- Contango, spot price < far month price or near month price < far month price
- Backwardation, spot price > far month price or near month price > far month price

2. Crypto Derivatives: Futures, Cross Exchange Spread



- Cross exchange arbitrage strategy can also be applied to crypto futures.
- Future rollover trading can be executed across exchanges chasing better spread.

Option

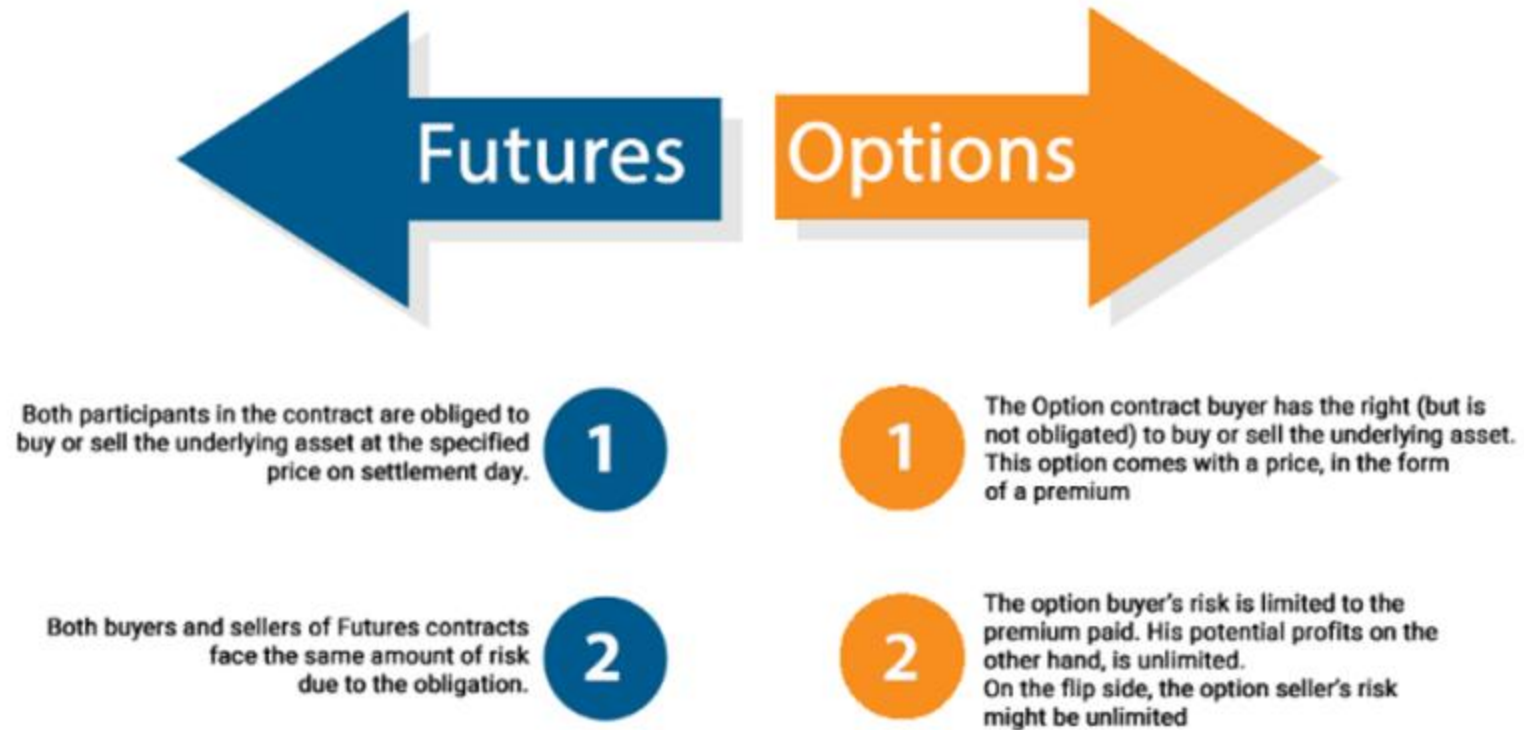
2. Crypto Derivatives: Option

Options contracts are agreements between a buyer and seller which give the buyer the right to buy or sell a particular asset at a later date (expiration date) and an agreed-upon price (strike price).



2. Crypto Derivatives: Option

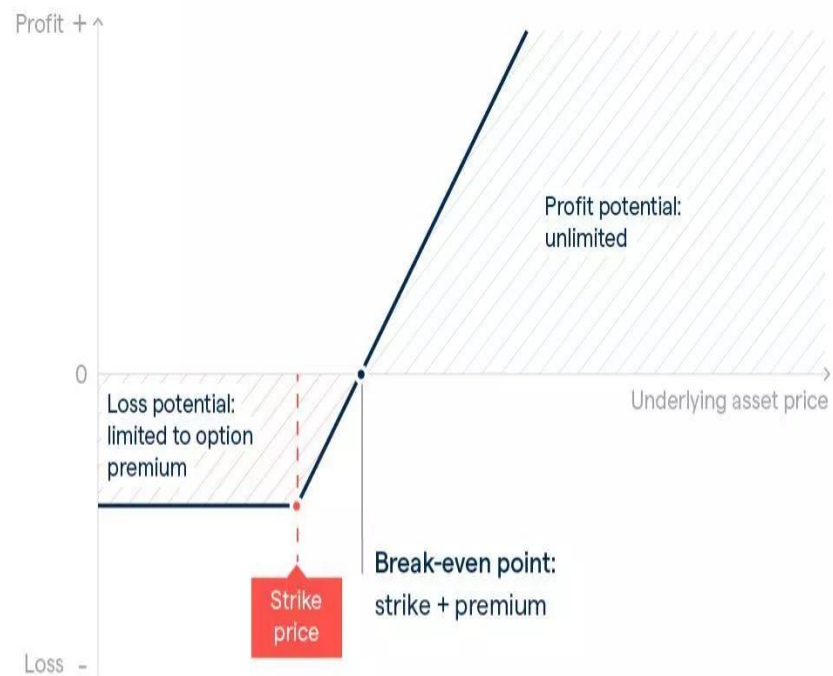
Futures vs Options



2. Crypto Derivatives: Option

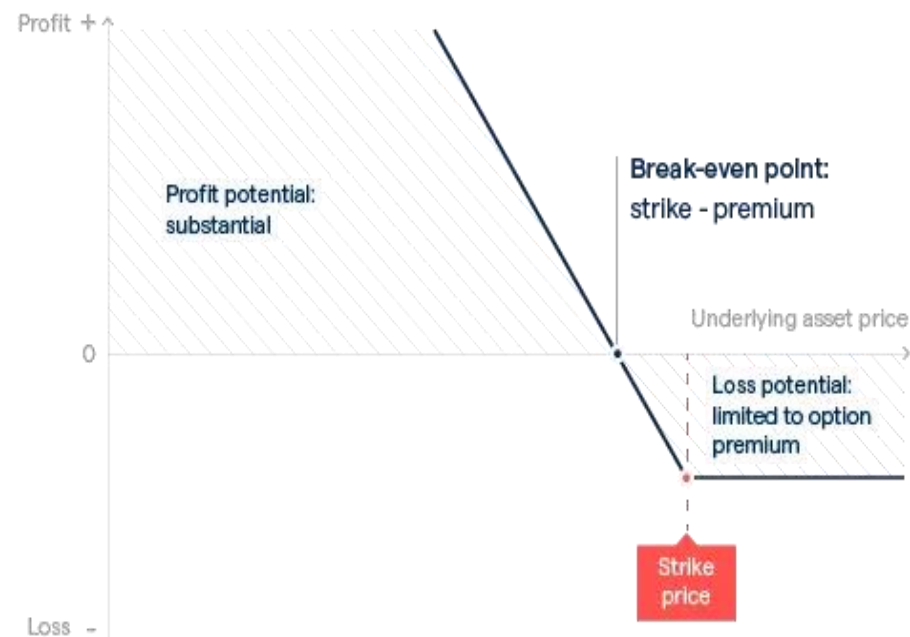
Buying call options

Buy a call option if you think the market will rise

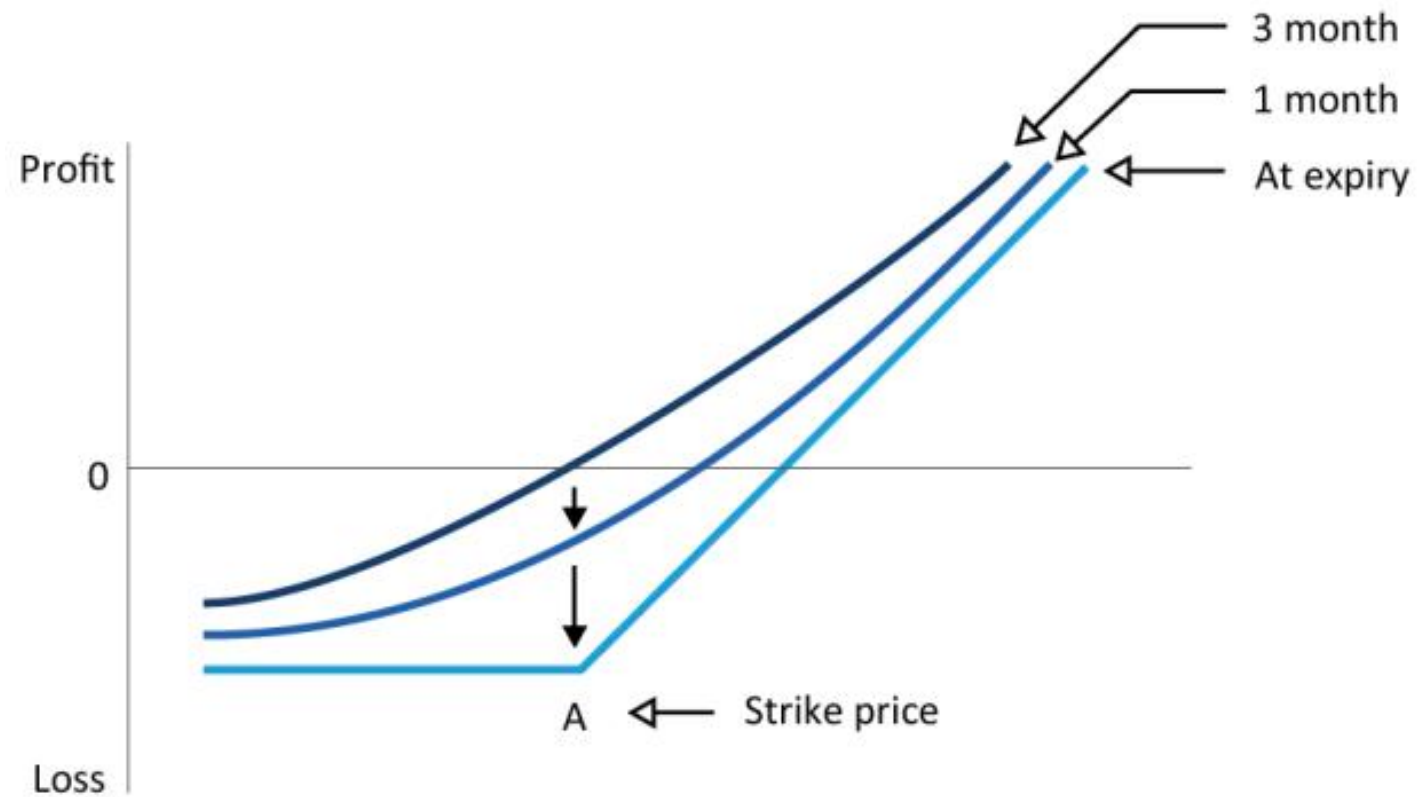


Buying put options

Buy a put option if you think the market will fall



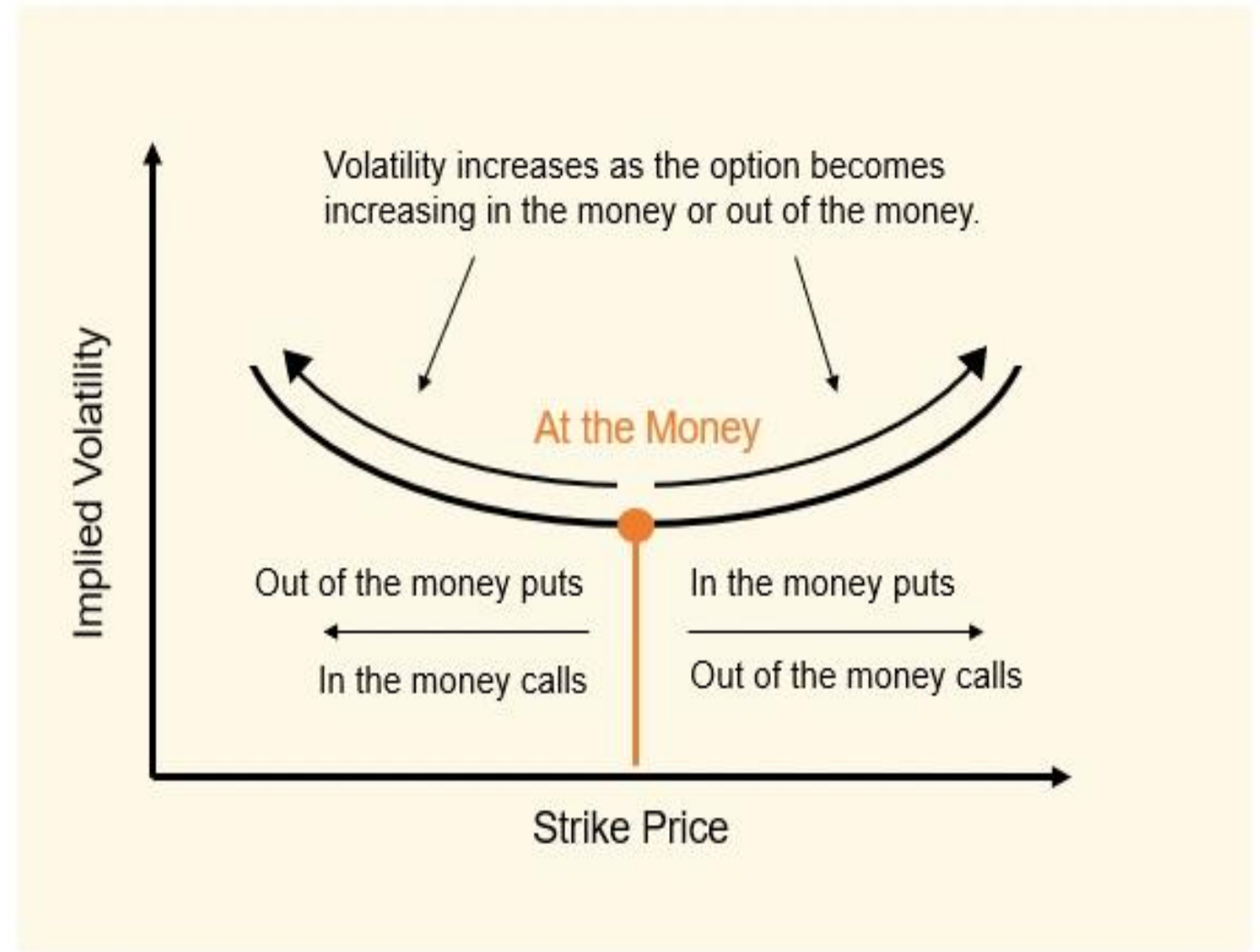
2. Crypto Derivatives: Option, Time Value of Option



2. Crypto Derivatives: Option, Volatility Smile

A volatility smile is a U-shaped curve showing that options far from the current price (deep in-the-money or out-of-the-money) have higher implied volatility (IV) than options close to the current price (at-the-money), when plotted against their strike prices for the same expiry date.

This pattern deviates from the flat volatility assumption of simple models like Black-Scholes, reflecting the market's expectation of greater risk for extreme price moves, especially after events like the 1987 crash.



2. Crypto Derivatives: Option, Volatility Smile, Observations

- An implied volatility smile means that Away-from-the-Money options have a higher implied volatility than At-the-Money options
- Implied volatility smile-shaped curves are typical of highly speculative markets
- Many researches on implied volatility curve dynamics showed that the 75% of all volatility changes consist of a shift, up or down, of the entire curve
- Smile-shaped curves are frequently found in equity index options, stock options and the most popular commodities / currencies
- Volatility smile curves can turn into a smirk if investors, traders and market players are expecting a market crash or if the plunge in price has already happened

2. Crypto Derivatives: Option, BTC Options

Deribit

SpotFuturesOptionsStrategy

\$88,52746.83F/8h: +0.000%

Log InRegister

Options (BTC)

ALL15 DEC 2516 DEC 2517 DEC 2518 DEC 2519 DEC 2526 DEC 2502 JAN 2630 JAN 2627 FEB 2627 MAR 2626 JUN 2625 SEP 26

ColumnsAround ATMStrike RangeExpected price moveUSDDist

CSV

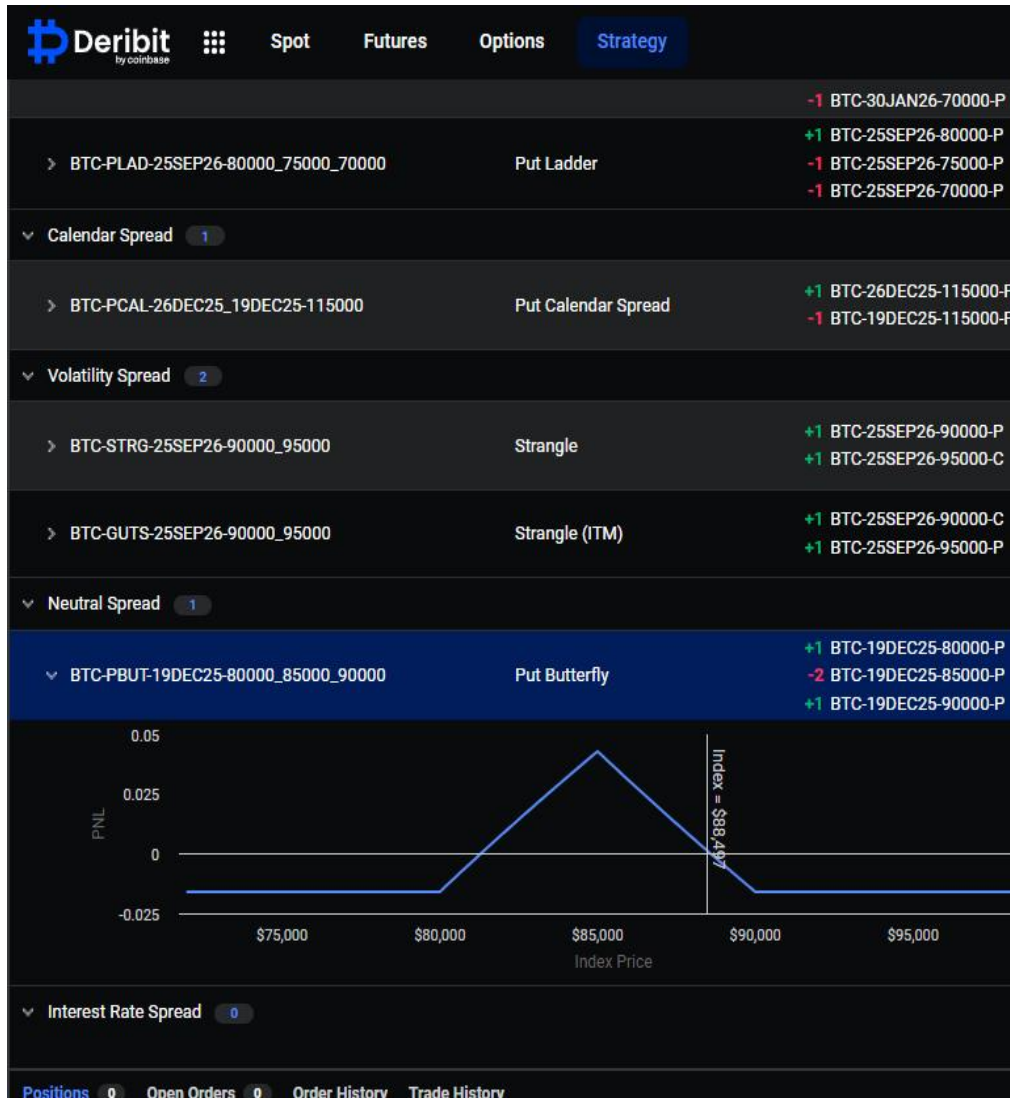
Calls										Puts									
Underlying future: \$ 88,536.00										IV: 54.8% (-1,624, +1,654)									
Open	ΔDelta	Size	IV Bid	Bid	Mark	Ask	IV Ask	Size	Strike	Size	IV Bid	Bid	Mark	Ask	IV Ask	Size	ΔDelta	Open	
-	0.98	0.1	-	0.0360 \$3187.20	0.0514 72.90%	0.0895 \$7923.72	441.5%	9.6	84,000	52.6	66.6%	0.0001 \$8.85	0.0002 72.90%	0.0003 \$26.56	78.6%	5.6	-0.02	200.5	
-	0.91	4.0	-	0.0240 \$2124.80	0.0296 62.56%	0.0650 \$5754.66	356.1%	4.0	86,000	8.0	61.6%	0.0009 \$79.68	0.0010 62.56%	0.0011 \$97.39	65.0%	7.5	-0.09	143.5	
6.9	0.81	3.0	52.0%	0.0185 \$1637.86	0.0195 57.76%	0.0210 \$1859.20	71.9%	2.4	87,000	8.2	56.9%	0.0021 \$185.92	0.0022 57.76%	0.0024 \$212.48	60.0%	1.6	-0.19	176.1	
2.1	0.63	0.2	50.4%	0.0105 \$929.60	0.0112 55.62%	0.0115 \$1018.13	61.6%	14.7	88,000	0.1	54.5%	0.0050 \$442.67	0.0051 55.62%	0.0055 \$486.93	58.3%	9.6	-0.37	136.3	
15.3	0.40	2.8	51.9%	0.0050 \$442.67	0.0053 54.05%	0.0055 \$486.93	55.6%	1.6	89,000	22.3	50.1%	0.0100 \$885.33	0.0105 54.05%	0.0115 \$1018.13	57.5%	4.6	-0.60	367.0	
475.8	0.20	2.7	53.3%	0.0020 \$177.07	0.0021 54.43%	0.0023 \$203.63	57.3%	1.2	90,000	2.3	41.4%	0.0175 \$1549.33	0.0186 54.43%	0.0195 \$1726.40	62.7%	0.2	-0.80	189.1	
141.3	0.08	10.0	53.1%	0.0006 \$53.12	0.0007 55.27%	0.0009 \$79.68	58.6%	1.5	91,000	0.1	-	0.0275 \$2434.74	0.0285 55.27%	0.0295 \$2611.81	70.4%	0.1	-0.92	23.4	
223.8	0.04	4.0	60.1%	0.0003 \$26.56	0.0003 61.27%	0.0004 \$35.41	63.4%	9.0	92,000	0.1	-	0.0385 \$3408.53	0.0395 61.27%	0.0405 \$3585.60	83.8%	0.1	-0.96	52.1	
179.5	0.02	13.1	62.4%	0.0001 \$8.85	0.0001 64.71%	0.0002 \$17.71	69.0%	12.6	93,000	26.1	-	0.0495 \$4382.39	0.0505 64.71%	0.0520 \$4603.73	95.0%	0.1	-0.98	20.3	
140.9	0.00	-	-	-	0.0000 65.06%	0.0002 \$17.71	81.5%	6.2	94,000	0.1	-	0.0210 \$1859.20	0.0617 65.06%	0.1035 \$9163.19	471.2%	0.1	-1.00	0.7	
55.4	0.00	-	-	-	0.0000 65.07%	0.0001 \$8.85	85.2%	4.1	95,000	0.1	-	0.0720 \$6374.39	0.0730 65.07%	0.0745 \$6595.72	123.1%	0.1	-1.00	2.2	
31.0	0.00	-	-	-	0.0000 65.07%	0.0001 \$8.85	96.1%	7.2	96,000	5.0	-	0.0001 \$8.85	0.0843 65.07%	0.1270 \$11243.72	530.7%	0.1	-1.00	-	
11.2	0.00	-	-	-	0.0000 65.07%	0.0001 \$8.85	117.3%	6.8	98,000	5.0	-	0.0001 \$8.85	0.1069 65.07%	0.1515 \$13412.78	594.6%	0.1	-1.00	0.4	
28.0	0.00	-	-	-	0.0000 65.07%	0.0001 \$8.85	137.5%	13.4	100,000	-	-	-	0.1295 65.07%	-	-	-	-1.00	-	

Positions0Open Orders0Order HistoryTrade HistoryExpiry GreeksAccount Summary

SearchAllBTCETHUSDCAllFuturesOptionsNo groupingCurrencyExpiry

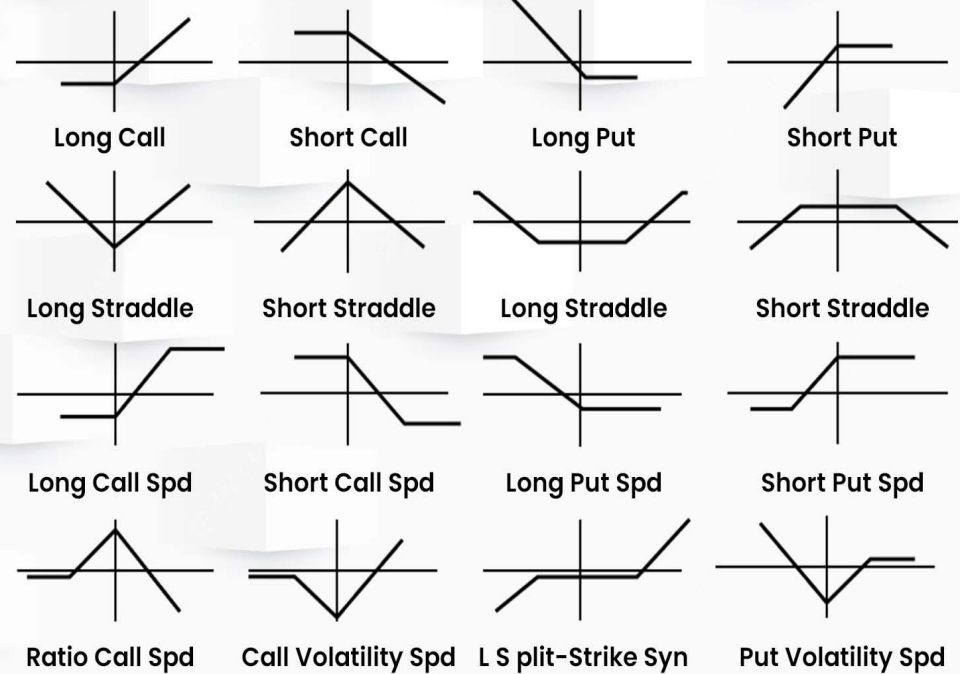
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2. Crypto Derivatives: Option, Option Combo Strategies



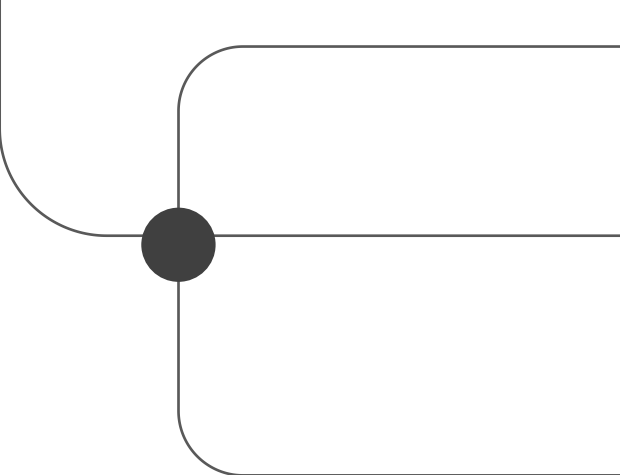
Different Option Strategies That Investors Should Know

Some Options Strategy Payoffs

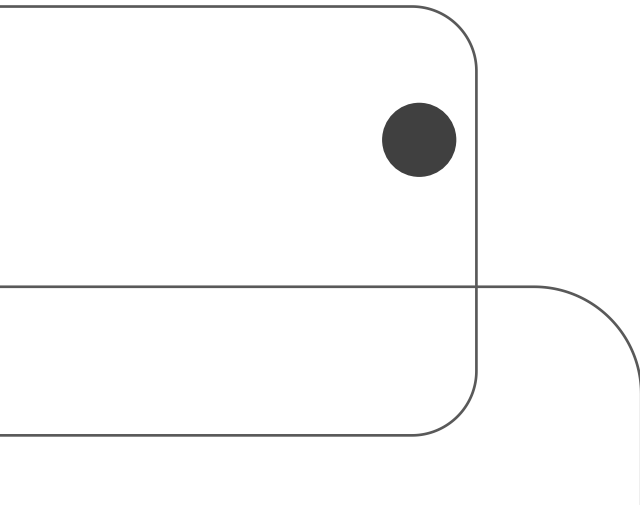


Strike

more info: <https://www.strike.money/options/option-trading-strategies>



3. Alternative Crypto Markets

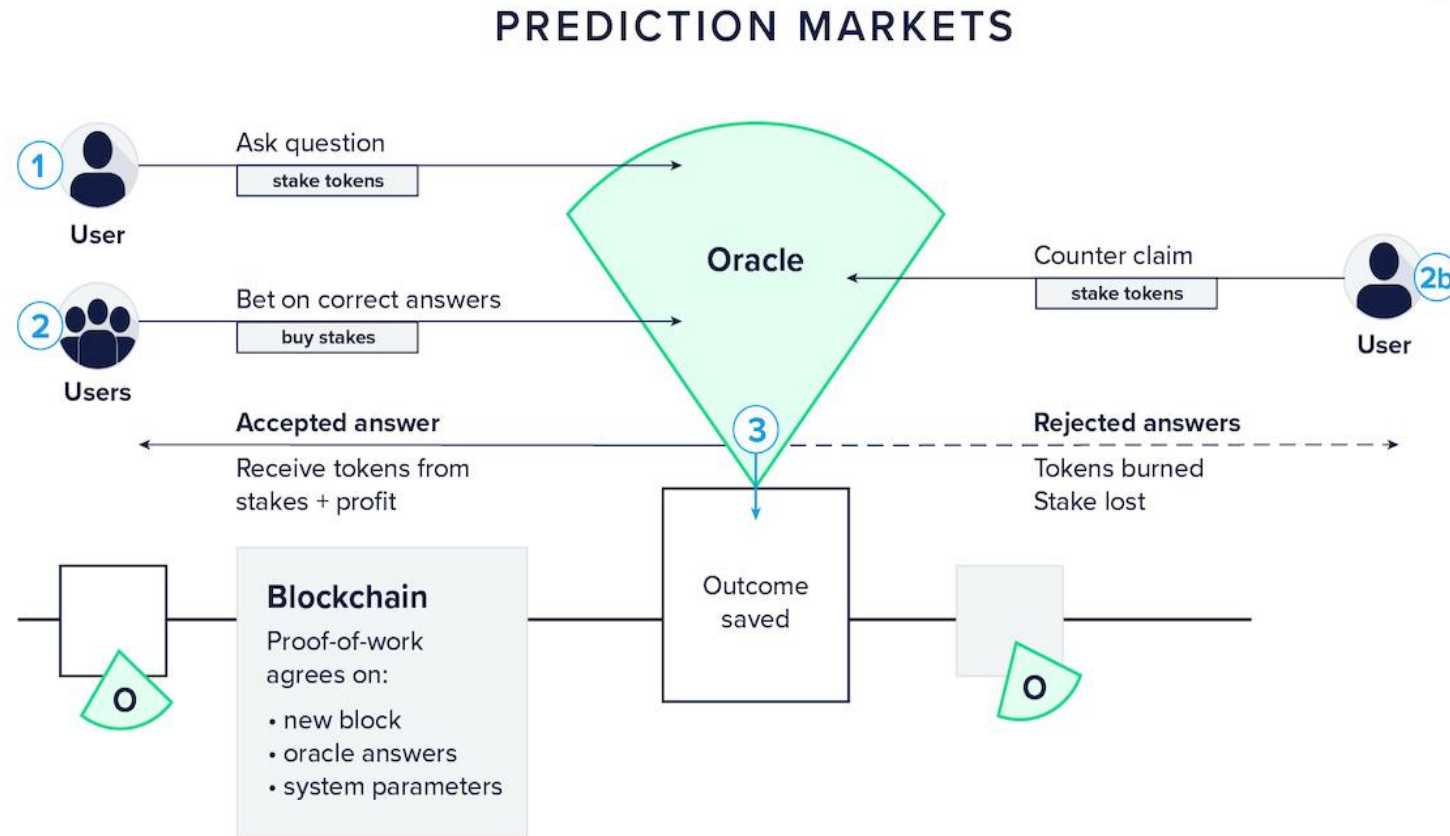


3. Alternative Crypto Markets: Prediction Market, Betting on Real World Events



- “Prediction market” is where you can bet on real world events.
- “Yes” or “No”: The outcome has to be binary.
- Winner takes all; loser loses all; Bookmaker charges small.

3. Alternative Crypto Markets: Prediction Market, Smart Contract and Blockchain



- Participants deposit tokens into a smart contract, which processes win–loss settlement and distributes payouts triggered by Oracle updates.
- The smart contract ensures the security, fairness, and transparency of the betting process.

3. Alternative Crypto Markets: Prediction Market, Polymarket

The screenshot displays the Polymarket website interface. At the top, there is a navigation bar with the Polymarket logo, a search bar, and links for 'How it works', 'Log In', and 'Sign Up'. Below the navigation bar, a horizontal menu lists various categories: Trending, Breaking, New, Politics, Sports, Finance, Crypto, Geopolitics, Earnings, Tech, Culture, World, Economy, Elections, Mentions, and More. A secondary search bar and filter tabs (All, Trump, Fed, Syria, Venezuela, Ukraine, Warner Bros, Best of 2025, Epstein, Equities, Weather, Derivatives) are located below the menu. The main content area is a grid of prediction market cards. Each card features a title, a question, a list of outcomes with their respective probabilities, and a 'Yes'/'No' button. The cards are organized into sections: 'Fed decision in January?', 'Who will Trump nominate as Fed Chair?', 'Maduro out by...?', 'U.S. forces seize another Venezuela-linked oil shi...', 'Packers vs Broncos', 'Colts vs Seahawks', 'Hornets vs Cavaliers', 'Flyers vs Hurricanes', 'Bucks vs Nets', 'Chile Presidential Election', 'How many Gold Cards will Trump sell in 2025?', and 'Thailand x Cambodia ceasefire by...?'. Each card also displays the volume of bets (Vol.) and a 'LIVE' status indicator.

Fed decision in January?

Outcome	Probability
50+ bps decrease	1%
25 bps decrease	21%

Who will Trump nominate as Fed Chair?

Outcome	Probability
Kevin Hassett	52%
Kevin Warsh	39%

Maduro out by...?

Outcome	Probability
December 31, 2025	8%
January 31, 2026	19%

U.S. forces seize another Venezuela-linked oil shi...

Outcome	Probability
Yes	18%
No	

Packers vs Broncos

Team	Probability
Packers	53%
Broncos	48%

Colts vs Seahawks

Team	Probability
Colts	47%
Seahawks	54%

Hornets vs Cavaliers

Team	Probability
Hornets	56%
Cavaliers	45%

Flyers vs Hurricanes

Team	Probability
Flyers	9%
Hurricanes	92%

Bucks vs Nets

Team	Probability
Bucks	52%
Nets	49%

Chile Presidential Election

Outcome	Probability
José Antonio Kast	100%
Jeannette Jara	<1%

How many Gold Cards will Trump sell in 2025?

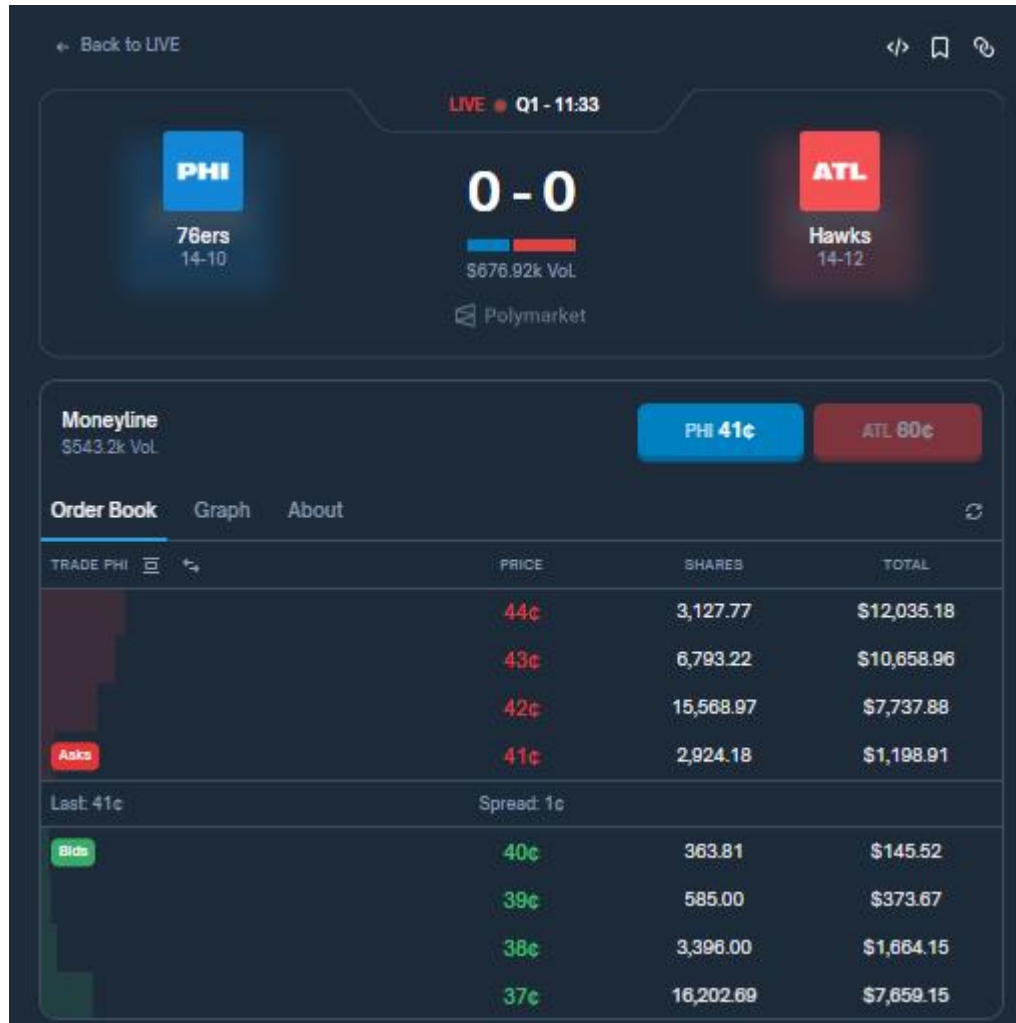
Outcome	Probability
0	91%
1-100	8%

Thailand x Cambodia ceasefire by...?

Outcome	Probability
December 15	2%
December 31	36%

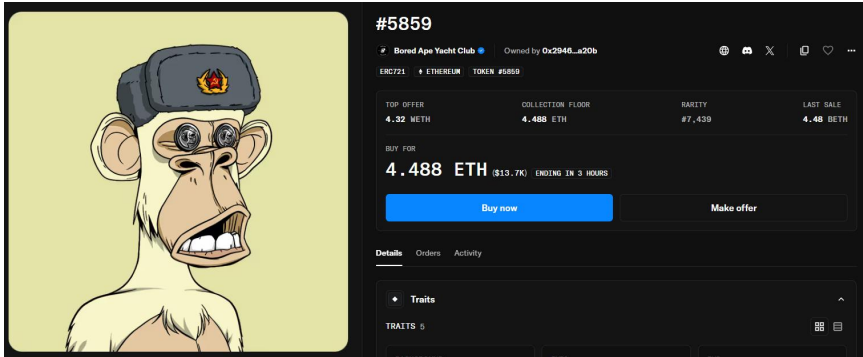
- “Prediction market” is where you can bet on real world events.
- Polymarket is a legally compliant platform regulated by the CFTC, with approval granted in November 2025.

3. Alternative Crypto Markets: Prediction Market, Polymarket

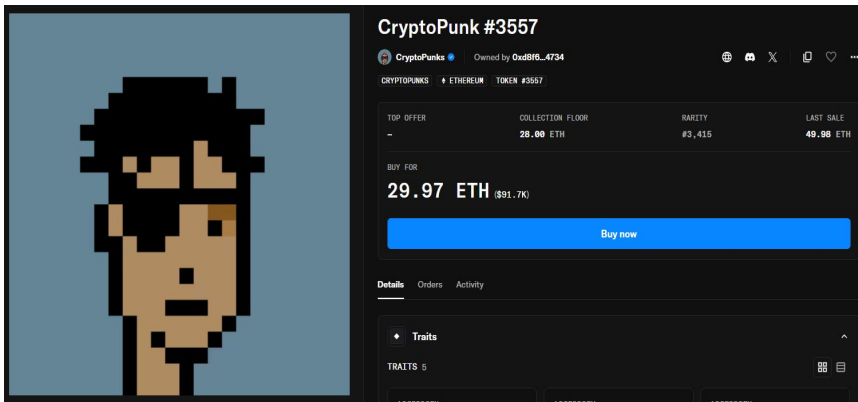


- Polymarket is now commission free.
- It is a good chance to be a “market maker”, a “trader”, or a “arbitrager” instead of a “gambler”.

3. Alternative Crypto Markets: Non-Fungible Token (NFT)



2 year full time: A mid-level professional ANNUAL income ranges from \$2400-7200 in Myanmar.



Brand New 2026 Cayenne, \$105,470

- An NFT (Non-Fungible Token) is a unique, blockchain-based digital certificate that establishes verifiable ownership and provenance of a specific asset, whether digital or physical.
- In short, An NFT is a token with a unique ID.

3. Alternative Crypto Markets: Non-Fungible Token (NFT), OpenSea

The screenshot displays the OpenSea marketplace interface. On the left, there's a sidebar with navigation options like 'Collections', 'Tokens', and 'Chains'. The main area shows a list of NFT collections with columns for 'COLLECTION', 'FLOOR PRICE', '1D CHANGE', 'TOP OFFER', '1D VOL', '1D SALES', 'OWNERS', and 'SUPPLY'. A modal window is open for the 'CryptoPunks' collection, showing its floor price, 1D change, 24h volume, and items count, along with a preview of the NFTs.

COLLECTION	FLOOR PRICE	1D CHANGE	TOP OFFER	1D VOL	1D SALES	OWNERS	SUPPLY
Bored Ape Yacht Club	4.488 ETH	-1.8%	4.41 WETH	116.06 ETH	24	5,498	9,911
Pudgy Penguins	4.58 ETH	-4.6%	4.36 WETH	92.34 ETH	20	4,990	8,888
Hypurr	497.00 HYPE	-3.5%	485.00 WHYPE	3,025.04 HYPE	16	3,737	4,600
Lil Pudgys	0.5885 ETH	-3.5%	0.578 WETH	49.52 ETH	84	10,173	21,999
CryptoPunks	28.00 ETH	-6%	-	26.25 ETH	1	3,789	9,999
XCOPY editions 2019-22	2.299 ETH	-0%	-	10.00 ETH	1	707	1,400
Doodles	0.517 ETH	-6%	0.507 WETH	8.14 ETH	15	4,435	9,999
fwogs	0.0703 ETH	+3.4%	0.059 WETH	7.63 ETH	97	1,841	5,555

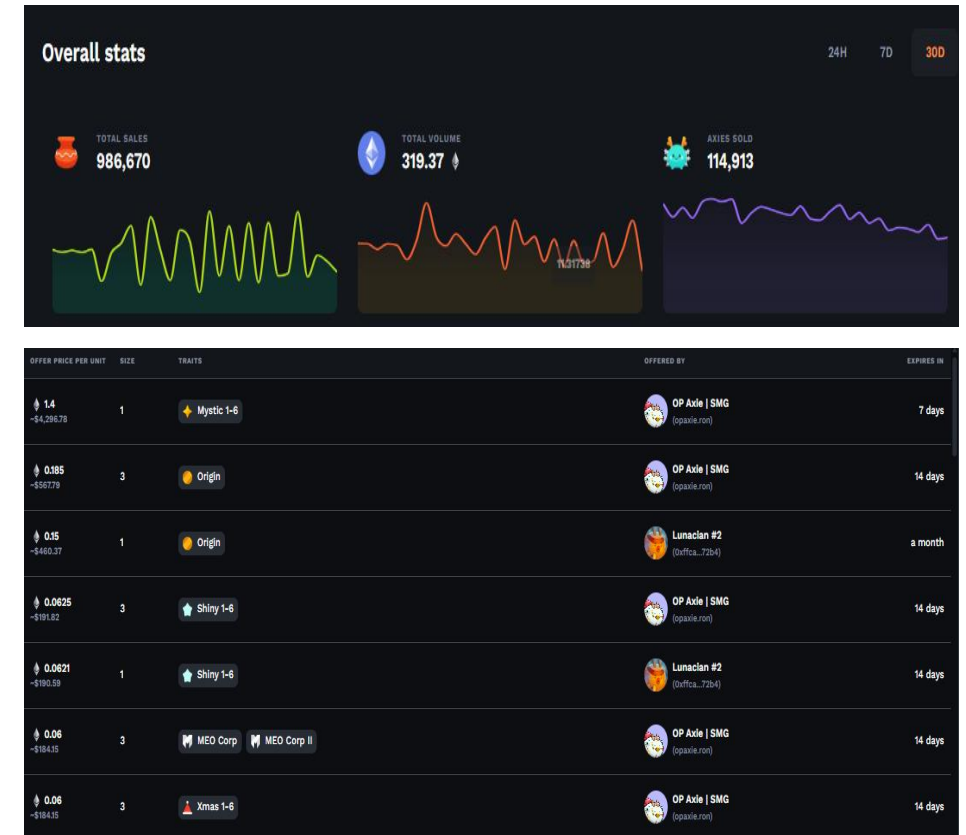
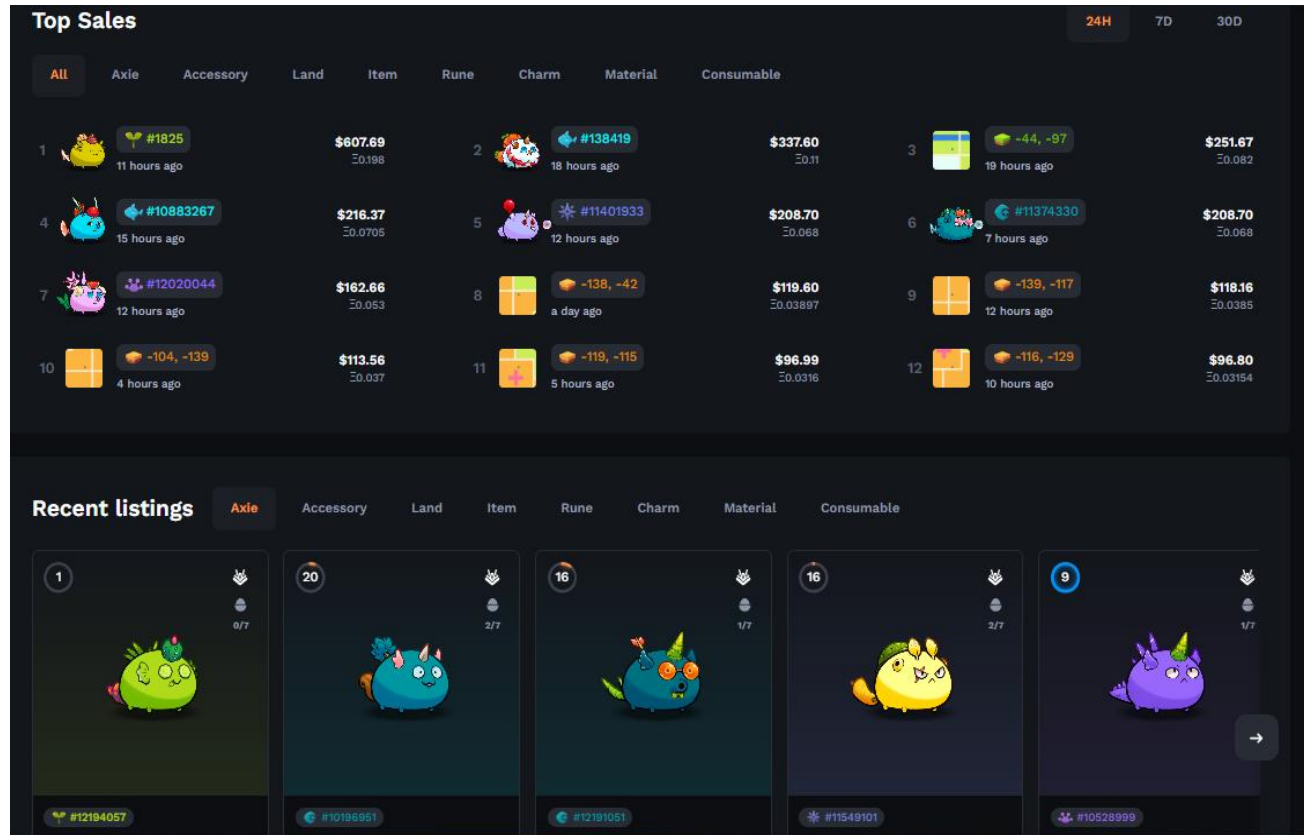
- OpenSea is a leading marketplace for NFTs.
- NFTs are significantly less liquid than Bitcoin.

3. Alternative Crypto Markets: GameFi, Axie Infinity



- GameFi (Game Finance) combines blockchain, DeFi, and online gaming to enable play-to-earn (P2E) models, where players earn tradable NFTs and tokens with real-world value, giving them ownership and income potential beyond the game itself.

3. Alternative Crypto Markets: GameFi, Axie Infinity, Marketplace



- In Axie Infinity, in-game assets (characters, equipment, land, etc.) are represented as NFTs and can be freely traded for ETH.
- 30-day volume is near \$1 million.
- Try NFT market making? Check: Lommers, K., Kim, J., & Baioumy, M. (2022). Market Making in NFTs. Available at SSRN 4226987.

Assignment 1 (Regular Track): Crypto Market Analysis

Lecture 1

- 1.(10) Download data from <https://coinmarketcap.com/historical/> by hand OR through APIs from any crypto exchange you prefer (e.g. coinbase, binance, etc.).
- 2.(10) Check and download data from <https://www.blockchain.com/explorer/charts> , either by hand or through API is fine.
- 3.(30) List the data you think that can be used to predict BTC prices and explain the reasons behind each. Quantitative explanations are preferred.

Lecture 2

- 4.(10) Find and implement a limit order book simulator on Github. Display and explain its outcomes.

Lecture 3

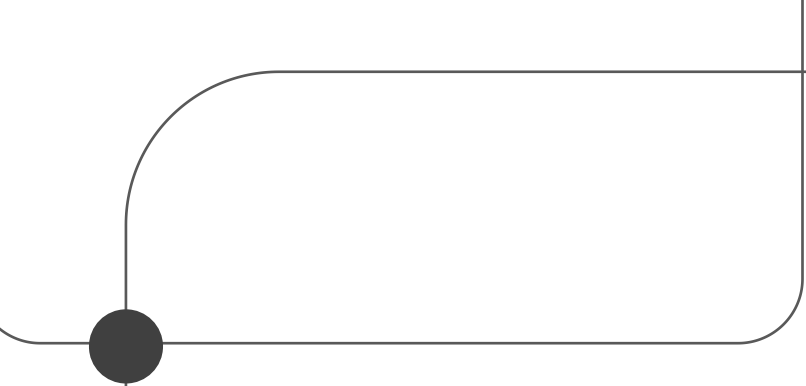
- 5.(10) Locate the crypto asset(s) you want to trade. Describe its statistical features, risk factors, and other information you think is important.
- 6.(30) What kind of strategies do you think could be applied to the asset(s) you selected? Explain how it is related to your strategy.

Assignment 1 (Express Track): Crypto Market Analysis

Challenge 1(100): Implement the factor model from the paper: Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. *The Journal of Finance*, 77(2), 1133-1177.

The paper's factor model excludes blockchain-specific variables because it is designed as a general asset-pricing framework rather than a BTC-focused predictor. Can you combine both blockchain factors and price factors together for BTC and ETH? This problem serves as preparation for designing market-neutral strategies.

Challenge 2(100): Write a simulator for automated market maker (AMM) from Uniswap. Compare the price/liquidity generated from the traditional simulator against the AMM. The purpose of this problem is to prepare for developing a cross-exchange arbitrage strategy.



Thank You

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